

1. GNM

GNM is a statistical model comprising a linear identity and expression basis, a skeletal rig for neck and eyeballs articulation, joint location identity basis and standard linear blend skinning (LBS) for posing the mesh vertices. The model conceptually follows a standard 3D Morphable Model (3DMM) definition (Blaiz and Vetter, 1999; Li et al., 2017)

1.1. Model Formulation

GNM is formulated as a function $\mathcal{M}(\Theta; \Psi) : \mathbb{R}^{|\Theta|} \rightarrow \mathbb{R}^{N_V \times 3}$, which produces a human head mesh of N_V 3D vertices given a set of model parameters Θ and fixed model data Ψ . The model parameters $\Theta = (\beta, \phi, \theta, \tau)$ consist of identity $\beta \in \mathbb{R}^{|\beta|}$ parameters, expression $\phi \in \mathbb{R}^{|\phi|}$ parameters, angle-axis rotations of $K = 4$ joints $\theta \in \mathbb{R}^{K \times 3}$ and global translation $\tau \in \mathbb{R}^3$. The model data $\Psi = (\mathbf{T}, \mathbf{J}, \mathbf{I}, \mathbf{E}, \mathbf{Q}, \mathbf{W}, \mathbf{p})$ consist of a template head mesh $\mathbf{T} \in \mathbb{R}^{N_V \times 3}$, template joint locations $\mathbf{J} \in \mathbb{R}^{K \times 3}$, identity basis $\mathbf{I} \in \mathbb{R}^{|\beta| \times 3 \times N_V}$, expression basis $\mathbf{E} \in \mathbb{R}^{|\phi| \times 3 \times N_V}$, joint location identity basis $\mathbf{Q} \in \mathbb{R}^{|\beta| \times 3 \times K}$, LBS weights $\mathbf{W} \in \mathbb{R}^{K \times N_V}$ and the kinematic chain definition $\mathbf{p} \in \mathbb{Z}^K$.

The model function is defined as $\mathcal{M}(\Theta; \Psi) = \mathcal{L}(\mathbf{V}_B, \mathbf{X}; \mathbf{W})$, where $\mathcal{L}$ is a standard LBS function, which rotates and blends the bind pose vertices $\mathbf{V}_B \in \mathbb{R}^{N_V \times 3}$ by the skinning transformations $\mathbf{X} \in \mathbb{R}^{K \times 4 \times 4}$ and blendweights $\mathbf{W}$. The bind pose vertices are computed as $\mathbf{V}_B = \mathcal{T}(\beta, \phi; \mathbf{T}, \mathbf{I}, \mathbf{E})$, where the function $\mathcal{T} : \mathbb{R}^{|\beta| \times |\phi|} \rightarrow \mathbb{R}^{N_V \times 3}$ applies per-vertex identity and expression offsets to the template $\mathbf{T}$. Formally,

$$\mathcal{T}(\beta, \phi; \mathbf{T}, \mathbf{I}, \mathbf{E}) = \mathbf{T} + \sum_i^{|\beta|} \beta_i \mathbf{I}_i + \sum_i^{|\phi|} \phi_i \mathbf{E}_i. \quad (1)$$

The skinning transforms are computed as $\mathbf{X} = \mathcal{X}(\mathbf{J}, \theta, \tau; \mathbf{p})$, where the function $\mathcal{X} : \mathbb{R}^{|\mathbf{J}| \times |\theta| \times 3} \rightarrow \mathbb{R}^{K \times 4 \times 4}$, takes the bind pose joint locations $\mathbf{J} \in \mathbb{R}^{K \times 3}$ and propagates the global translation τ and per-joint rotations θ through the kinematic chain $\mathbf{p}$ to obtain global per-joint affine transforms.

The bind pose joint locations are computed as $\mathbf{J} = \mathcal{J}(\beta; \mathbf{J}, \mathbf{Q})$, where the function $\mathcal{J} : \mathbb{R}^{|\beta|} \rightarrow \mathbb{R}^{K \times 3}$ is defined as

$$\mathcal{J}(\beta; \mathbf{J}, \mathbf{Q}) = \mathbf{J} + \sum_i^{|\beta|} \beta_i \mathbf{Q}_i. \quad (2)$$

Finally, the i -th output mesh vertex $\mathbf{v}^{(i)} \in \mathbb{R}^3$ is computed as

$$\mathcal{L}^{(i)}(\mathbf{V}_B, \mathbf{X}; \mathbf{W}) = \sum_{k=1}^K \mathcal{H}^{-1}(\mathbf{W}_{k,i} \mathbf{X}_k \mathcal{H}(\mathbf{V}_{B_i}^\top)), \quad (3)$$

where $\mathcal{H} : \mathbb{R}^3 \rightarrow \mathbb{R}^4$, $\mathcal{H}(\mathbf{v}) = \begin{pmatrix} \mathbf{v} \\ 1 \end{pmatrix}$ converts a 3D vector into homogeneous coordinates, and $\mathcal{H}^{-1}$ does the opposite.

The template head mesh $\mathbf{T}$, identity basis $\mathbf{I}$, expression basis $\mathbf{E}$ and joint location basis $\mathbf{Q}$ are learned from the data, whereas the skinning blendweights $\mathbf{W}$ and the kinematic chain $\mathbf{p}$ are designed manually by an artist, see Figure 1. The template joint locations $\mathbf{J}$ rely both on an artist defined joint regressor and the data driven identity basis $\mathbf{I}$.

1.2. Composite Linear Bases

To achieve higher fidelity of the shape space covered by the model, and to allow for the use of highly-specialized datasets, we split the identity and expression bases $\mathbf{I}$ and $\mathbf{E}$ into portions which

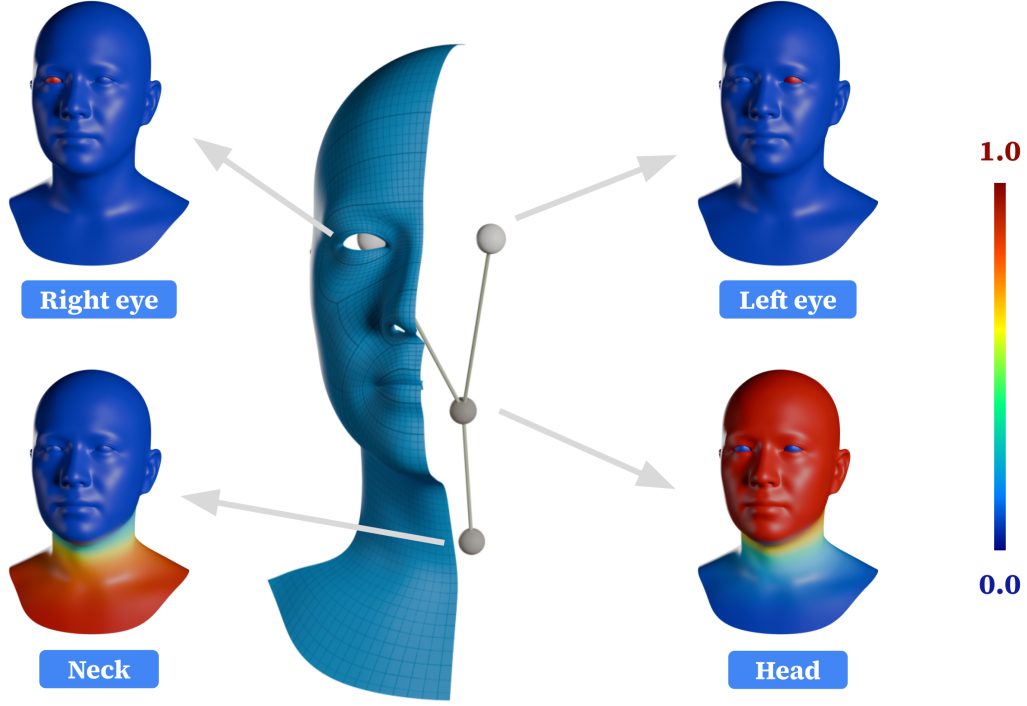


Figure 1 | Template mesh and skinning blendweights. The central model illustrates the neutral head template. The surrounding heatmaps display the manually designed blendweights for the four articulated joints: the right eye, left eye, neck, and head. The colour scale indicates the degree of joint influence over the mesh, ranging from 0.0 (blue, no influence) to 1.0 (red, full influence).

correspond to anatomical regions of the human head.

Specifically,

$$\mathbf{I} = \mathbf{I}^{(\text{head})} \parallel \mathbf{I}^{(\text{eye})} \parallel \mathbf{I}^{(\text{teeth})}, \quad (4)$$

where $\mathbf{I}^{(\text{head})}, \mathbf{I}^{(\text{eye})}, \mathbf{I}^{(\text{teeth})} \in \mathbb{R}^{|\beta^{(r)}| \times N_V \times 3}$ are the head, teeth and eyeballs portions of $\mathbf{I}$ stacked along the first tensor dimension, and where $|\beta^{(r)}|$ represents the number of components in portion $r \in \{\text{head}, \text{eye}, \text{teeth}\}$. Note, that $\beta = [\beta^{(\text{head})^\top} \beta^{(\text{eye})^\top} \beta^{(\text{teeth})^\top}]^\top$.

Similarly,

$$\mathbf{E} = \mathbf{E}^{(\text{left eye})} \parallel \mathbf{E}^{(\text{right eye})} \parallel \mathbf{E}^{(\text{lower face})} \parallel \mathbf{E}^{(\text{tongue})} \parallel \mathbf{E}^{(\text{pupil})}, \quad (5)$$

where $\mathbf{E}^{(\text{left eye})}, \mathbf{E}^{(\text{right eye})}, \mathbf{E}^{(\text{lower face})}, \mathbf{E}^{(\text{tongue})}, \mathbf{E}^{(\text{pupil})} \in \mathbb{R}^{|\phi^{(r)}| \times N_V \times 3}$ are the left and right periocular region, lower face, tongue and eyeball pupil portions of $\mathbf{E}$, and where $|\phi^{(r)}|$ represents the number of components in portion $r \in \{\text{left eye}, \text{right eye}, \text{lower face}, \text{tongue}, \text{pupil}\}$. Note, that $\phi = [\phi^{(\text{left eye})^\top} \phi^{(\text{right eye})^\top} \phi^{(\text{lower face})^\top} \phi^{(\text{tongue})^\top} \phi^{(\text{pupil})^\top}]^\top$.

The definition of each of the portions of $\mathbf{I}$ and $\mathbf{E}$ is detailed in the following sections.

1.3. Head Identity

Here we define the head identity basis $\mathbf{I}^{(\text{head})}$, the template mesh $\mathbf{T}$, the template joint locations $\mathbf{J}$ and the joint location identity basis $\mathbf{Q}$. The head identity basis $\mathbf{I}^{(\text{head})}$ is computed as a standard PCA

over the set of neutral face meshes. As $\mathbf{I}^{(\text{head})}$ should only capture the deformation due to the change of human identity and not due to a rigid head movement in space, we first rigidly align all the neutral face meshes to a common template using Procrustes alignment (Luo and Hancock, 2002), obtaining the dataset $\widehat{\mathbf{X}}_{\mathbf{N}} \in \mathbb{R}^{N_N \times 3N_V}$ of N_N samples representing flattened 3D mesh vertices.

While $\widehat{\mathbf{X}}_{\mathbf{N}}$ contains all N_V vertices of the GNM topology, we zero-out the vertices corresponding to the eyeballs, and place them in the final model manually to achieve a higher fidelity of the eyeball-eyelid contact. We denote this modified dataset $\mathbf{X}_{\mathbf{N}}$. First, we explain how we compute the identity basis $\mathbf{I}_{\mathbf{s}}^{(\text{head})} \in \mathbb{R}^{|\beta| \times N_V \times 3}$ which is equivalent to $\mathbf{I}^{(\text{head})}$ except for the zeroed-out eyeball vertices. Next, we explain how we inject the eyeball deformation caused by the identity change into $\mathbf{I}_{\mathbf{s}}^{(\text{head})}$ to obtain the final $\mathbf{I}^{(\text{head})}$.

We compute $\bar{\mathbf{I}}, \mathbf{\Lambda} = \text{eig}(\mathbf{C}_{\mathbf{I}})$, where $\bar{\mathbf{I}} \in \mathbb{R}^{N_I \times 3N_V}$ and $\mathbf{\Lambda} \in \mathbb{R}^{N_I}$ are the eigenvectors and their associated eigenvalues and $\mathbf{C}_{\mathbf{I}} \in \mathbb{R}^{3N_V \times 3N_V}$ is a covariance matrix of the centered dataset $\bar{\mathbf{X}}_{\mathbf{N}} = \mathbf{X}_{\mathbf{N}} - \mathbf{1}\bar{\mathbf{x}}_{\mathbf{N}}$, where $\bar{\mathbf{x}}_{\mathbf{N}} = \frac{1}{N_N} \sum_{i=1}^{N_N} \mathbf{X}_{\mathbf{N}i}$ is the dataset mean and $\mathbf{1}$ is a vector of ones. Let $\mathcal{F} : \mathbb{R}^{a \times b} \rightarrow \mathbb{R}^{ab}$ be a function that flattens a matrix to a vector, while $\mathcal{F}^{-1}$ does the opposite. The i -th component of $\mathbf{I}_{\mathbf{s}}^{(\text{head})}$ is computed as $\mathbf{I}_{\mathbf{s}_i}^{(\text{head})} = \mathcal{F}^{-1}(\bar{\mathbf{I}}_i \mathbf{\Lambda}_i)$, that is, each basis vector is scaled proportionally to the dataset variance it explains, which unifies the effective range of the identity coefficients β . The basis vectors are thus orthogonal but not unit-length. Note, that we only keep the first $|\beta^{(\text{head})}| \ll N_I$ basis vectors which jointly explain $\sim 99\%$ of the dataset variance.

We express the GNM head template mesh as $\mathbf{T} = \mathcal{F}^{-1}(\bar{\mathbf{x}}_{\mathbf{N}})$, which we further modify by replacing the teeth by the mean of the teeth dataset $\mathbf{X}_{\mathbf{T}}$ of Section 1.5.2, and by optimizing eyeball location to best fit the eyelids.

To backfill the eyeballs in the identity basis, we optimize a uniform rigid translation for the eyeball vertices in each basis component to maintain a plausible eyeball-eyelid contact under identity deformations. For each basis component $\mathbf{I}_{\mathbf{s}_i}^{(\text{head})}$, we consider the positive and negative bound deformations of the eyelid vertices at a fixed magnitude and find an optimal translational offset such that the displaced eyeballs best fit the corresponding eyelid vertices of the positive and negative deformations. Thus we ensure that as the face identity changes, the eyeballs translate to stay aligned with the deforming eyelids. This modification results in the final $\mathbf{I}^{(\text{head})}$.

The template joints $\mathbf{J}$ were placed in the template mesh $\mathbf{T}$ manually by an artist, whereas the joint location identity basis $\mathbf{Q}$, which ensures that the skeleton scales and moves appropriately with each subject’s unique head shape controlled by the identity parameters β , is computed automatically using a linear joint regressor $\mathbf{R} \in \mathbb{R}^{K \times N_V}$. Specifically, the i -th joint component $\mathbf{Q}_i = \mathbf{R}\mathbf{I}_i$. The joint regressor itself is computed as an optimization problem $\mathbf{R} = \arg\min_{\mathbf{R}} \|\mathbf{R}\mathbf{T} - \mathbf{J}\|_{\text{Frob}} + \mathcal{L}_{\text{reg}}$, where $\mathcal{L}_{\text{reg}}$ is a regularizer which encourages sparsity and left-right mesh symmetry.

1.4. Head Expression

Here we define the left and right periocular region and the lower face portions $\mathbf{E}^{(\text{left eye})}$, $\mathbf{E}^{(\text{right eye})}$ and $\mathbf{E}^{(\text{lower face})}$ of $\mathbf{E}$. Each region is computed as an uncentered PCA over the set of samples representing vertex displacements of expressive faces w.r.t. the neutral one for each human subject. Similarly to $\mathbf{I}$, $\mathbf{E}$ should only capture shape variation due to a facial expression change, but not any rigid face motion. Therefore, we first align the registered meshes using a face mesh stabilization technique described in more detail at the end of this section.

Let $\mathbf{V}_{\mathbf{n}}^{(s)}, \mathbf{V}_{\mathbf{e}_i}^{(s)} \in \mathbb{R}^{N_V \times 3}$ be the neutral and the i -th expressive face mesh of subject s , where $\mathbf{V}_{\mathbf{e}_i}^{(s)}$ is stabilized to $\mathbf{V}_{\mathbf{n}}^{(s)}$, and let $\boldsymbol{\delta}_i^{(s)} \in \mathbb{R}^{3N_V}$, $\boldsymbol{\delta}_i^{(s)} = \mathcal{F}(\mathbf{V}_{\mathbf{e}_i}^{(s)} - \mathbf{V}_{\mathbf{n}}^{(s)})$ represent a corresponding flattened array

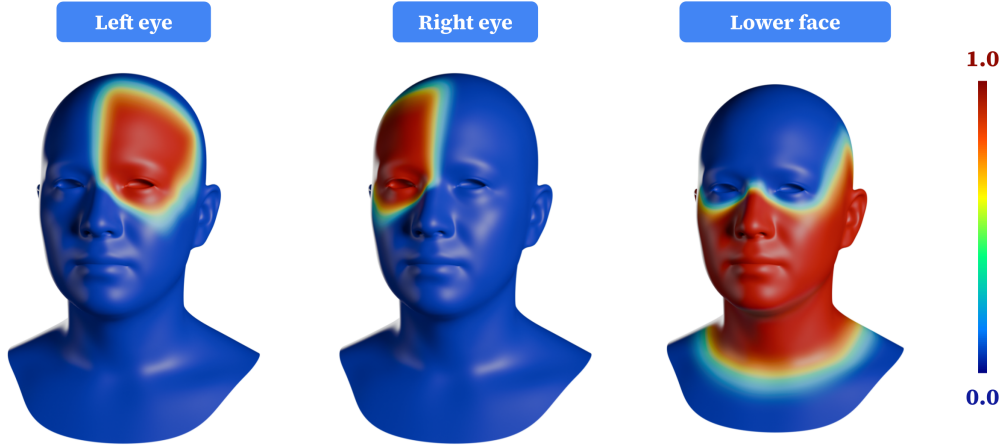


Figure 2 | Regional expression basis masks. Visualisation of the continuous vertex masks (S_r) used to spatially partition the head mesh into three distinct areas: the left periocular region, the right periocular region, and the lower face. Warmer colours indicate higher mask weights ($S_r \rightarrow 1$). This regional formulation isolates the computation of the expression basis, yielding two critical benefits: it provides localised, intuitive control (preventing semantic leakage, such as a jaw movement triggering an eye wink), and it ensures the left and right periocular expressions can be perfectly mirrored.

of per-vertex deltas. Let N_M be the total number of dataset samples, N_s be the total number of unique subjects and n_s be the number of expressive faces available for a subject s . The expression dataset $\mathbf{X}_E \in \mathbb{R}^{N_M \times 3N_V}$, is then defined as $\mathbf{X}_E = [\boldsymbol{\delta}_1^{(1)} \dots \boldsymbol{\delta}_{n_1}^{(1)} \boldsymbol{\delta}_1^{(2)} \dots \boldsymbol{\delta}_{n_2}^{(2)} \dots \boldsymbol{\delta}_1^{(N_s)} \dots \boldsymbol{\delta}_{n_{N_s}}^{(N_s)}]^\top$. Any deformation of the tongue or eyeballs is zeroed-out, as we compose their expression basis individually, see Sections 1.5.2 and 1.5.3. However, note that the motion of the jaw together with the lower teeth is modeled fully within $\mathbf{E}$.

Dividing the head mesh into the left and right periocular regions and the lower face region (see Figure 2) for the sake of computing the expression basis comes with two benefits: (i) the model allows for localized and more intuitive control of a facial expression (e.g. an opening mouth will not trigger an eye wink), and (ii) the left and right periocular region expressions are perfectly mirrored. Let $S_r \in \mathbb{R}^{3N_V}$ be a vertex mask of a region r with values in $[0, 1]$. We then define a per-region expression dataset $\mathbf{X}_E^{(r)}$, composed of samples $\mathbf{X}_{E_i}^{(r)} = \mathbf{X}_{E_i} S_r$.

We compute $\bar{\mathbf{E}}^{(r)}, \Pi^{(r)} = \text{eig}(\mathbf{C}_E^{(r)})$, where $\bar{\mathbf{E}}^{(r)} \in \mathbb{R}^{N_r \times 3N_V}$ and $\Pi^{(r)} \in \mathbb{R}^{N_r}$ are the eigenvectors and their associated eigenvalues, N_r is the number of eigenvectors computed for region r , and $\mathbf{C}_E^{(r)} \in \mathbb{R}^{3N_V \times 3N_V}$ is a covariance matrix of *uncentered* dataset $\mathbf{X}_E^{(r)}$. We avoid centering $\mathbf{X}_E^{(r)}$ since modifying the template mesh $\mathbf{T}$ to absorb the mean of $\mathbf{X}_E^{(r)}$ would result in a face with slightly closed eyes and a slightly opened mouth. Instead, to make controlling GNM intuitive, we design the model so that zero identity and expression parameters $\boldsymbol{\beta}, \boldsymbol{\phi}$ produce a perfectly neutral face.

The i -th component is computed as $\mathbf{E}_i^{(r)} = \mathcal{F}^{-1}(\bar{\mathbf{E}}_i^{(r)} \Pi_i^{(r)})$ so as to scale the components proportionally to the explained variance, and we keep only the first $|\boldsymbol{\phi}^{(r)}| \ll N_r$ which jointly explain $\sim 99\%$ of the dataset variance. The expression components $\mathbf{E}^{(r)}$ are orthogonal within each region r , but not among the regions, as the vertex masks of the neighboring regions have small overlaps, where the contribution of the per-region components are linearly blended. Note, that we only compute $\mathbf{E}^{(\text{left_eye})}$, and then mirror the individual displacement vectors along a vertical plan splitting the head into symmetric left and right halves, to obtain $\mathbf{E}^{(\text{right_eye})}$.

As mentioned above, to remove any spurious misalignment within the source dataset, we first stabilize the face meshes. Stabilization finds a rigid 6-DOF transform between a source and a target mesh representing a skin surface of the same human subject, so that the underlying (and unknown) skull aligns perfectly in space. It is a notoriously difficult vision problem (Beeler and Bradley, 2014; Wu et al., 2018), therefore, to achieve a precise alignment, we take a two stage approach.

First, we apply fully automatic modified confidence map stabilization (Bednarik et al., 2024). Then, any remaining misalignments are removed by a semi-automatic PCA-based stabilization, which operates as follows. We define five granular facial regions (left and right periocular region, nose, mouth, neck with back of the head), and perform a PCA decomposition for each, conceptually taking the same steps as when computing $\mathbf{E}^{(r)}$. The assumption is that any strong spurious rigid 6-DOF deformation would get naturally contained in a few PCA components. We visualize the contribution of the individual per-region components, a human operator visually identifies and discards the ones representing an unwanted deformation, and we rebuild the data from the remaining ones, thus creating a clean set of expressive meshes to build $\mathbf{X}_E$ from.

1.5. Internal Anatomy

To make GNM truly comprehensive, internal anatomy must be included too. Modeling the teeth, tongue and eyeballs poses a significant challenge due to the data scarcity, complex teeth geometry, and a high degree of freedom in the motion of a tongue. We address these limitations by leveraging a hybrid approach that combines artist-guided synthetic models and diverse real-world 3D scans.

1.5.1. Teeth modeling

GNM integrates a parametric dental subsystem that replaces static, generic mouth templates a shape controlled by the teeth identity basis $\mathbf{I}^{(\text{teeth})}$ and the corresponding parameters $\boldsymbol{\beta}^{(\text{teeth})}$. The model is designed to represent diverse dental arches, accurately capturing natural variations in teeth shape, size, and individual alignment while remaining compatible with the overall head shape.

Starting from a generic template teeth model, an artist rigged each tooth and procedurally generated $N_T = 5000$ synthetic dental shapes, including both the upper and lower teeth along with their associated gums. We denote this dataset $\mathbf{X}_T \in \mathbb{R}^{N_T \times 3N_V}$. Note that all the vertices except for teeth are set to 0.

As before, $\mathbf{I}^{(\text{teeth})}$ is computed as a standard PCA on the dataset $\mathbf{X}_T$, where the eigenvectors are scaled by their respective eigenvalues represent the final $\mathbf{I}^{(\text{teeth})}$. We again only keep the first $|\boldsymbol{\beta}^{(\text{teeth})}|$ components which explain $\sim 99\%$ of the dataset variance.

1.5.2. Tongue modeling and articulation

We develop a tongue model to address the lack of expressiveness in the inner-cavity in traditional 3DMMs. The tongue model is represented as the portion $\mathbf{E}^{(\text{tongue})}$ of the expression basis $\mathbf{E}$, and it is, too, computed as a PCA decomposition of a dataset of vertex displacements w.r.t. the neutral template.

The tongue represents a highly deformable surface which is notoriously difficult to register from raw scans (Ploumpis et al., 2021). To capture the broad tongue shape space, we generate a specialized dataset by fitting a custom artist-made tongue rig to two data sources. First, we take a subset of expressive faces performing specific tongue movements (sideway motion, rolling etc.) from our registrations, we only keep the outer skin part of the face and use our tongue rig to generate varied synthetic tongue poses by respecting the overall face and lip geometry to avoid any penetrations.

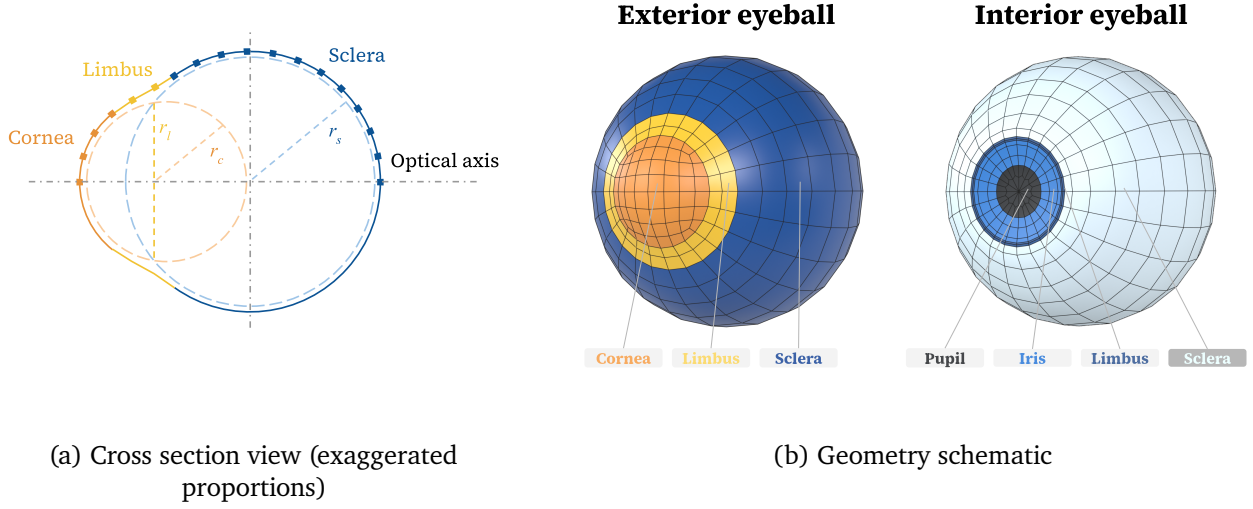


Figure 3 | Parametric representation of the GNM ocular geometry. (a) A 2D cross-sectional view illustrating the underlying two-sphere model. The optical axis connects the centres of the scleral and corneal spheres. To maintain anatomical compatibility with the eyelids, the scleral radius (r_s) is held constant, whilst the corneal (r_c) and limbal (r_l) radii are sampled from physiological distributions. (b) 3D schematics of the exterior and interior eyeball meshes. The linear identity basis ($\mathbf{I}^{(\text{eye})}$), derived from the 2D cross-section polylines, is interpolated onto these 3D surfaces as a surface of revolution. The interior mesh highlights the pupil, which is independently driven by a dedicated expression basis ($\mathbf{E}^{(\text{pupil})}$) to enable dynamic, lighting-dependent dilation.

Second, we fit the tongue rig to the dataset of sparse 3D keypoints from (Medina et al., 2022) while applying rigorous constraints to the tongue’s global morphology and curvature to ensure anatomical plausibility.

As in Section 1.4, we compute per-vertex displacements to the template, and denote the resulting dataset as $\mathbf{X}_G \in \mathbb{R}^{N_G \times 3N_V}$, where $N_G \sim 2.5K$ is the dataset size. Note, that all the vertices except for the tongue are set to 0. As before, $\mathbf{E}^{(\text{tongue})}$ is computed as a standard (centered) PCA on the dataset $\mathbf{X}_G$, where the eigenvectors $\bar{\mathbf{E}}^{(\text{tongue})}$ scaled by their respective eigenvalues $\Pi^{(\text{tongue})}$ represent the components of $\mathbf{E}^{(\text{tongue})}$. As before, we only keep the first $|\phi^{(\text{tongue})}| \ll N_{\text{tongue}}$ components explaining $\sim 99\%$ of the dataset variance.

Note, that the mean $\bar{\mathbf{x}}_G$ of the dataset $\mathbf{X}_G$ must be included as a summand in Eq. 1 producing the posed head vertices. An intuitive solution would be to absorb $\bar{\mathbf{x}}_G$ in $\mathbf{T}$. However, as $\mathbf{E}^{(\text{tongue})}$ is computed independently of the rest of the face, $\bar{\mathbf{x}}_G$ represents a tongue protruding the surface of a neutral face mesh. Therefore, we instead absorb $\bar{\mathbf{x}}_G$ as the first component of $\mathbf{E}^{(\text{tongue})}$. Setting ϕ to the default value of $\mathbf{0}$ thus leads to a retracted tongue tucked away inside the mouth cavity.

1.5.3. Eyeball Model Formulation

We define the eyeball as a linear parametric model designed to bridge the gap between simplified spherical representations and the complex, person-specific ocular geometry required for high-fidelity gaze tracking. The model represents the $\mathbf{I}^{(\text{eye})}$ portion of the identity basis $\mathbf{I}$ and the $\mathbf{E}^{(\text{pupil})}$ portion of the expression basis $\mathbf{E}$.

Accurate cornea modeling is particularly critical for synthetic data generation, as the subtle curvature of the cornea dictates the formation of LED glints, which are primary features in gaze-target

prediction (Guestrin and Eizenman, 2006). By integrating corneal and limbal parameters directly into the identity space $\mathbf{I}$, we ensure that every generated identity possesses biologically plausible ocular features compatible with the overall facial topology.

The eyeball geometry is defined as a two-sphere model where a large scleral sphere of radius r_s forms the bulk of the eye, and a smaller corneal sphere of radius r_c defines the anterior segment, see Figure 3a. The optical axis is established by the line connecting these two sphere centers. To capture human anatomical diversity while maintaining eyelid compatibility with the GNM template, the scleral radius is held constant ($r_s \approx 14.6$ mm). Conversely, the remaining parameters are drawn from established physiological Gaussian distributions—specifically for the limbus radius r_l ($\mu = 6.0$ mm, $\sigma = 0.44$ mm) and cornea radius r_c ($\mu = 8.5$ mm, $\sigma = 0.73$ mm)—to yield a pseudo-ground truth dataset.

To transform these non-linear geometric properties into a linear basis $\mathbf{I}^{(\text{eye})}$, we first sample a large set ($N = 10,000$) of 2D eyeball cross-sections as polylines. These polylines are generated by varying the nonlinear parameters (r_l , r_c) according to their physiological statistics. Importantly, a smoothing algorithm is applied to the transition zone between the sclera and cornea in each polyline to achieve a more organic and continuous surface, avoiding sharp geometric discontinuities. PCA is then applied to the vertices of these smoothed polylines to extract the leading modes of shape variation. The resulting 2D linear identity basis is subsequently transferred to the 3D template eyeball interior and exterior mesh vertices via interpolation in polar coordinates, exploiting the eyeball’s nature as a surface of revolution. This process allows the eyeball’s shape to be controlled by a small set of identity coefficients, seamlessly integrated into the primary GNM identity basis.

The eyeball expression basis $\mathbf{E}^{(\text{pupil})}$ is focused solely on pupil dilation. We employ a hand-crafted geometric basis parameterized such that a single coefficient, scaled to operate within a range of $[-3, 3]$, controls pupil dilation. At a coefficient value of -3, the pupil radius contracts to a point, at 0 it is half the radius of the iris, and at +3, it expands to match the full iris size. This range comfortably encompasses biologically sensible radii (e.g., 1 to 4 mm) and provides essential variability for lighting-dependent gaze models or expressive rendering. Both eyeballs of a GNM instance share the same shape identity $\mathbf{I}^{(\text{eye})}$ and expression $\mathbf{E}^{(\text{pupil})}$ parameters, ensuring bilateral symmetry. Anatomical coherence with the eyelids is primarily established during the GNM model construction phase: the main head identity basis components are analyzed for their impact on the eye socket shape, and corresponding rigid transformations are computed and integrated into both the vertex and joint identity bases for all eyeball vertices and their rotation pivots, ensuring the eyeballs translate appropriately with facial identity changes.

1.6. Initial Placement of Teeth and Tongue

The initial version of the GNM model had artist-designed teeth, tongue and eyeballs manually injected in the identity and expression basis, after it is computed from the registered data. Here we describe this process.

To backfill the teeth and tongue in the identity basis $\mathbf{I}$, we assume that any face deformation caused by an identity change can lead to a rigid 3-DOF translation of the teeth and tongue. As an approximation, we observe the mean displacement of the inner lower and upper lip vertices induced by each component $\mathbf{I}_i$ and store this value into the (thus far zeroed-out) portion of $\mathbf{I}_i$ corresponding to the teeth and tongue vertices.

To backfill the lower teeth and tongue in the expression basis $\mathbf{E}$, we first note that the any facial expression change which moves the jaw, rigidly transforms the lower teeth too. As an approximation, we define a vertex group covering the lower lip and the chin, and we find a rigid 6-DOF transform of

these vertices induced by each component $\mathbf{E}_i$ using a Procrustes alignment. The recovered transform is applied to all the lower teeth and tongue vertices in $\mathbf{E}_i$.

1.7. Model Implementation Details

As explained in section 1.2, the identity and expression bases $\mathbf{I}$ and $\mathbf{E}$ are composed of disjoint anatomical regions, each represented as a PCA over the vertex displacements, where we only keep the portion of the components to explain sufficient amount of the dataset variance. Table 1 summarizes the final number of components in each region. The topology of GNM contains $N_V = 17821$ vertices (including the outer skin, teeth, tongue and eyeballs) and the skeletal structure consists of $K = 4$ joints. The template head mesh $\mathbf{T}$ is placed in the global coordinate system with neck aligned upright along the positive Y-axis, while the face looks along the positive Z-axis.

Table 1 | The dimensions of the individual regions of the identity basis $\mathbf{I}$ and the expression basis $\mathbf{E}$.

Identity				Expression					
head	eye	teeth	TOTAL	left eye	right eye	lower face	tongue	pupil	TOTAL
170	3	80	253	100	100	150	31	1	382

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